MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST (SET – C)** | **Time:** 1 Hour
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided wherever specified.
2. **Section A** comprises Multiple Choice / Objective Questions of 1 mark each.
3. **Section B** comprises Short Answer Type questions.
4. **Section C** comprises Long Answer & Competency Based questions.
5. Use of calculators or electronic devices is strictly prohibited. Draw neat figures wherever required.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. If the distance between the points $A(4, p)$ and $B(1, 0)$ is 5 units, then the value of p is: [1]

(A) 4 only (B) ± 4 (C) -4 only (D) 0

Q2. The point which lies on the perpendicular bisector of the line segment joining the points $A(-2, -5)$ and $B(2, 5)$ is: [1]

(A) $(0, 0)$ (B) $(0, 2)$ (C) $(2, 0)$ (D) $(-2, 0)$

Q3. The distance between the points $(a\cos\theta + b\sin\theta, 0)$ and $(0, a\sin\theta - b\cos\theta)$ is: [1]

(A) $a^2 + b^2$ (B) $a + b$ (C) $a^2 - b^2$ (D) $\sqrt{a^2 + b^2}$

Q4. *Assertion [1]

(A) :* The point $(0, 4)$ lies on the y-axis. *Reason (R):* The x-coordinate of any point lying on the y-axis is always zero. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A). (C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q5. The distance between the points $A(0, 6)$ and $B(0, -2)$ is: [1]

(A) 6 (B) 8 (C) 4 (D) 2

SECTION B: Short Answer Type Questions (10 Marks)

Q6. Prove that abscissa of a point P which is equidistant from points with coordinates $A(7, 1)$ and $B(3, 5)$ is 2 more than its ordinate. [2]

OR Find the coordinates of a point on the line $x + y = 5$ which is equidistant from $(6, 4)$ and $(5, 2)$. [2]

SECTION B (Continued): Short Answer Type Questions (Questions 8 to 10)

Q8. Find the coordinates of a point P on x-axis equidistant from the points $A(-1, 0)$ and $B(5, 0)$, using distance formula. [2]

Q9. Write the distance of the point $(-3, 4)$ from the x-axis. [2]

Q10. If AOBC is rectangle whose vertices are $A(0, 3), O(0, 0), B(5, 0)$ and $C(5, 3)$, then find the length of its diagonal. [2]

SECTION C: Long Answer & Competency Based Questions (15 Marks)

Q11. Show that the points (a, a), $(-a, -a)$ and $(-\sqrt{3}a, \sqrt{3}a)$ are the vertices of an equilateral triangle. [5]

Q12. Find the ratio in which the point $P\left(\frac{3}{4}, \frac{5}{12}\right)$ divides the line segment joining the points $A\left(\frac{1}{2}, \frac{3}{2}\right)$ and $B(2, -5)$. [5]

Q13. In what ratio does the point $P(-4, 6)$ divide the line segment joining the points $A(-6, 10)$ and $B(3, -8)$? [5]

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